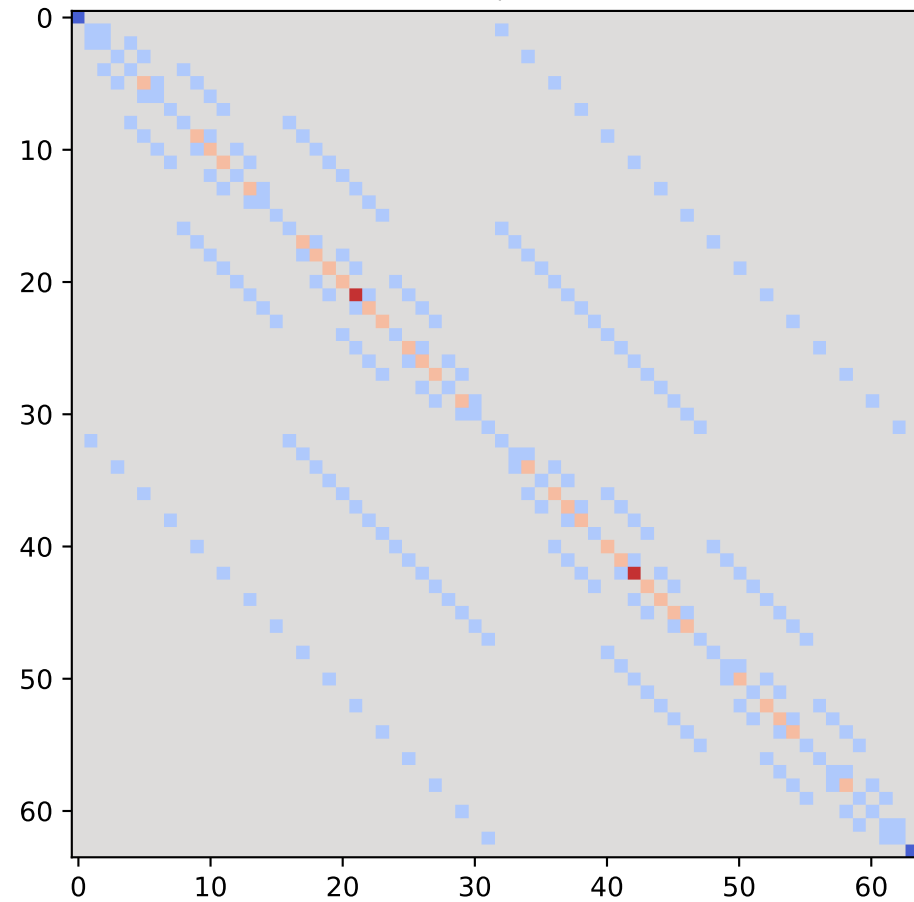
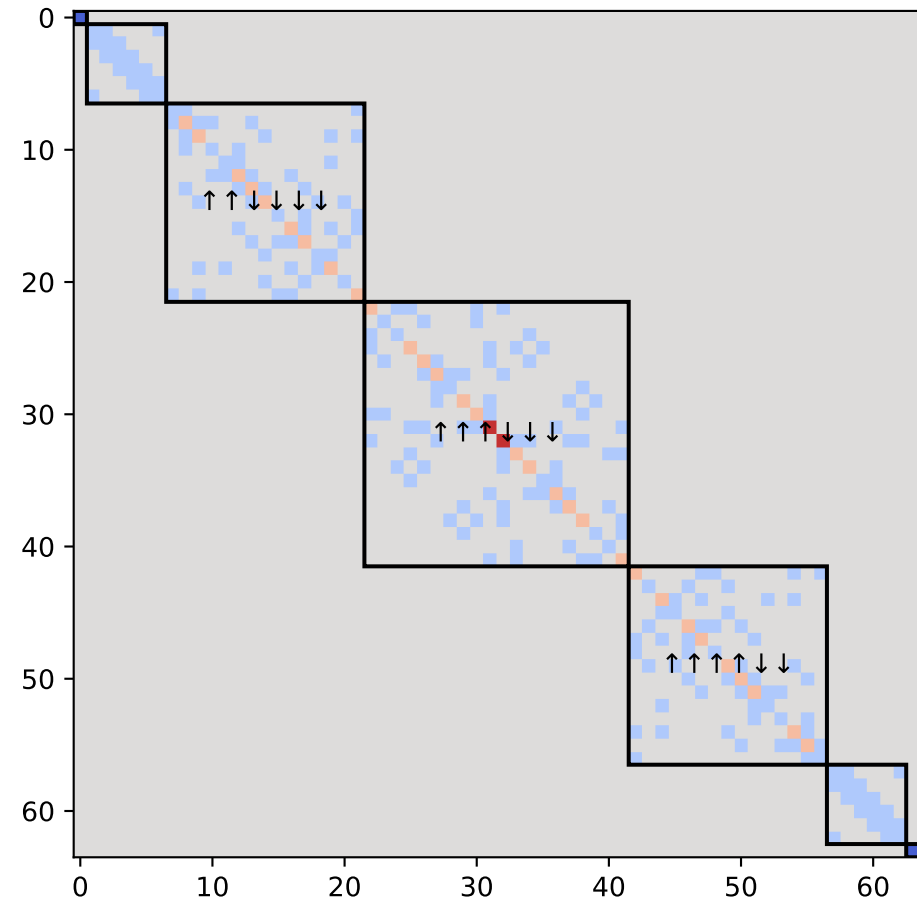


(i) Computational basis $\{|s_1 s_2 \cdots s_N\rangle, s_i \in \{\uparrow, \downarrow\}\}$



(ii) $U(1)$ Basis: Blocked by Magnetization S^z



$SU(2)$ basis: blocks labelled by (S, S^z)

